

7.1:

Let $P = \{p_0, p_1, \dots, p_{n-1}\}$

Where $p_0 = (0, 0)$ and for $i = 1, \dots, n-1$

$p_i = R(\cos((2\pi i)/(n-1)), \sin((2\pi i)/(n-1)))$ for $R > 0$

$p_1, \dots, p_{n-1}$ are the vertices of a regular $(n-1)$ sided polygon on a circle centered at origin, and p_0 is the center.

Point x is in the Voronoi cell of p_0 iff it is at least as close to p_0 as to ever p_i . Therefore:

$$V(p_0) = \bigcap_{(i=1, n-1)} h(p_0, p_i)$$

For fixed i , $\|x - p_0\| \leq \|x - p_i\|$

Since $p_0 = (0, 0)$, squaring both sides gives $\|x\|^2 \leq \|x - p_i\|^2 = \|x\|^2 - 2x * p_i + \|p_i\|^2$

So then, $2x * p_i \leq \|p_i\|^2 = R^2 \rightarrow x * p_i \leq R^2/2$

Hence,

$$V(p_0) = \bigcap_{(i=1, n-1)} \{x : x * p_i \leq R^2/2\}$$

Each inequality defines half-plane bounded by perpendicular bisector of segment $p_0 p_i$

Because vectors p_i are equally spaces around the circle, these bisectors are also equally spaced in direction. Intersection is regular $(n-1)$ sided polygon centered at the origin

A regular $(n-1)$ polygon has $n-1$ vertices. Therefore Veronoi cell $V(p_0)$ has $n-1$ vertices.

To conclude, for every $n > 3$, there exists a set of n sites in plane such that one Voronoi cell has $n-1$ vertices.

7.2:

Delaunay graph has one vertex for each site, and two sites are connected exactly when their Voronoi cells share an edge. So number of edges incident to site p_i in Delaunay graph is exactly the number of sides and therefore also number of vertices, of the Voronoi cell of p_i

So $v_i = \deg(p_i)$

Which means $1/n * \sum_{(i=1, n)} v_i = 1/n \sum_{(i=1, n)} \deg(p_i)$

Delaunay graph is planar, so triangulate every face. This only adds edges, so if E edges then

$$E \leq 3n - 3 - k$$

where k is number of hull vertices. Hence,

$$E \leq 3n - 6$$

Therefore the average degree is $1/n \sum_{(i=1, n)} \deg(p_i) = 2E/n \leq 2(3n - 6)/n = 6 - 12/n < 6$

Since $v_i = \deg(p_i)$, $1/n \sum_{(i=1, n)} v_i < 6$.

Therefore, the average number of vertices of a Voronoi cell is less than six.

7.5:

Single site parabola can show up in more than one disjoint arc on the beach line. Beach line is the lower envelope of all these parabolas, so one parabola can be lowest and disappear behind others before then becoming lowest again when farther to the side.

Beach line is balanced BST of arcs.

One site giving more than one arc example:

Imagine five sites where:

- One site p_i high above the others
- Two sites to the left and right a little below it
- And two more sites farther left and farther right, also below it

So can use $p_i = (0, 10)$, $a = (-1, 6)$, $b = (1, 6)$, $c = (-4, 2)$, and $d = (4, 2)$

For suitable position of sweep line below all these sites, parabola of p_i is cut by the parabolas of a , b , c , d so that along the beach line the order of arcs is: c , p_i , a , b , p_i , d

So the same site p_i contributes two arcs.

Beach line is not made from whole parabolas, but rather only the visible parts of them. Site's parabola may be visible in one interval of x -values is hidden in the middle by other sites and then is visible again later.

Therefore one site can contribute multiple disjoint arcs.

It can also contribute a linear number of arcs.

Putting one site p_i in a high position, and placing the other $n - 1$ sites below it in alternating left-right pattern, the parabola of p_i keeps reappearing between neighboring lower parabolas.

Then beach line looks like q_i , p_i , q_2 , p_i , q_3 , p_i , $\dots$, q_{n-1} , so p_i contributes $\Theta(n)$ arcs.

Therefore, there are configurations where one site contributes more than one arc, and in fact there are configurations where one site contributes a linear number of arcs, $\rightarrow \Theta(n)$

7.7:

Two sites $p_1 = (0, 3)$ and $p_2 = (1, 0)$

Sweep line is $y = \ell$, that moves down and decreases.

Break point is intersection of the two parabolas with focal points p_1 , p_2 , and directrix $y = \ell$,

Parabola Equations:

$$P_1: (x - 0)^2 + (y - 3)^2 = (y - \ell)^2$$

$$P_2: (x - 1)^2 + y^2 = (y - \ell)^2$$

Bisector Line:

$$\begin{aligned} (x - 0)^2 + (y - 3)^2 - (x - 1)^2 - y^2 &= 0 \\ \rightarrow 2x - 6y + 8 &= 0 \\ \rightarrow x &= 3y - 4 \end{aligned}$$

and every breakpoint lies on this line

Substitute $x = 3y - 4$ into p_2 parabola:

$$(3y - 5)^2 + y^2 = (y - \ell)^2$$

This simplifies to $9y^2 + (2\ell - 30)y + (25 - \ell^2) = 0$

Two breakpoints are the zeros of the equation.

$\ell = -1$, the zeros are about $y = 1.075$ and $y = 2.481$

$\ell = -2$, the roots are $y = 0.778$ and $y = 3$

As sweep line moves down from $\ell = -1$ to $\ell = -2$

- First breakpoint moves down: 1.075 to 0.778
- Second breakpoint moves up: 2.481 to 3

Therefore the breakpoints do not always move downwards.

7.10:

Algorithm:

1. Compute Delaunay triangulation $DT(P)$ of P in $O(n \log n)$ time
2. Check all edges of $DT(P)$ and compute its Euclidean distance length. Then return the shortest edge
3. Delaunay graph is planar, so only has $O(n)$ edges. Scan takes $O(n)$ time

Runtime: $O(n \log n) + O(n) = O(n \log n)$

Closest pair must be Delaunay edge:

p, q is closest pair in P , and $d = |pq|$

Closed disk has pq as the diameter.

Any point inside this disk is strictly closer than d to either p or q

But p, q is a closest pair, so no other point of P can lie inside this disk

Thus, there is a closed disk with p and q on its boundary and no other point of P in its interior.

Through the Delaunay property:

$pq \in DG(P)$ iff closed disk exists with p and q on boundary and no other site inside.

Therefore the closest pair p, q is an edge of the Delaunay graph

Because the closest pair is one of the Delaunay edges, taking the shortest Delaunay edge must return a closest pair. So algorithm is correct.

7.11:

Algorithm:

1. Compute the Voronoi diagram V or (P) of the n points in $O(n \log n)$ time. Compute Delaunay triangulation $DG(P)$ in $O(n \log n)$ time
2. For each point $p \in P$, look at all neighbors of p in the Delaunay graph
3. Among those neighbors, choose the one closest to p
4. Output that point as the nearest neighbor of p

Correctness:

If q is the nearest neighbor of p , then the Voronoi cells $V(p)$ and $V(q)$ share edge.

Also, p and q are adjacent in the Delaunay graph.

So point's nearest neighbor must appear as one of the Delaunay neighbors.

Algorithm checks all of these neighbors and picks the closest one, so it returns the correct nearest neighbor for every point.

Runtime:

Computing Delaunay triangulation takes $O(n \log n)$ time

The Delaunay graph is planar, so it has $O(n)$ edges total

Therefore the total # of neighbors that are checked over all points is $O(n)$

Total running time is $O(n \log n) + O(n) = O(n \log n)$